Chapter 25

Electric Potential

Sec 25.1: Electric Potential and Potential Difference

- When a test charge is placed in an electric field, it experiences a force
  - $\vec{\mathbf{F}} = q_o \vec{\mathbf{E}}$
  - The force is conservative
- If the test charge is moved in the field by some external agent, the work done by the field is the negative of the work done by the external agent
- The work done by the electric field is $\vec{\mathbf{F}} \cdot d = q_o \vec{\mathbf{E}} \cdot d$
- As this work is done by the field, the potential energy of the charge-field system is changed by $\Delta U = -q_o \vec{\mathbf{E}} \cdot d$
- The change in potential energy of the system

$$\Delta U = q\,\Delta V = -qEd$$

- The potential energy per unit charge, U/q_o, is the **electric potential**
  - The potential is characteristic of the field only
    - The potential energy is characteristic of the charge-field system
  - The potential is independent of the value of q_o
  - The potential has a value at every point in an electric field
- The electric potential is $V = \frac{U}{q_o}$
- The potential is a scalar quantity, Since energy is a scalar
- As a charged particle moves in an electric field, it will experience a change in potential $\Delta V = \frac{\Delta U}{q_o} = -\int_A^B \vec{\mathbf{E}} \cdot d$
- The difference in potential is the meaningful quantity
- We often take the value of the potential to be zero at some convenient point in the field
- Electric potential is a scalar characteristic of an electric field, independent of any charges that may be placed in the field
- Assume a charge moves in an electric field without any change in its kinetic energy

- The work performed on the charge is $W = \Delta U = q\,\Delta V$
- 1 V = 1 J/C
  - V is a volt
  - It takes one joule of work to move a 1-coulomb charge through a potential difference of 1 volt
- In addition, 1 N/C = 1 V/m
- This indicates we can interpret the electric field as a measure of the rate of change with the position of the electric potential with respect to position.
- Another unit of energy that is commonly used in atomic and nuclear physics is the electron-volt
- One electron-volt is defined as the energy a charge-field system gains or loses when a charge of magnitude e (an electron or a proton) is moved through a potential difference of 1 volt
  - $1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$

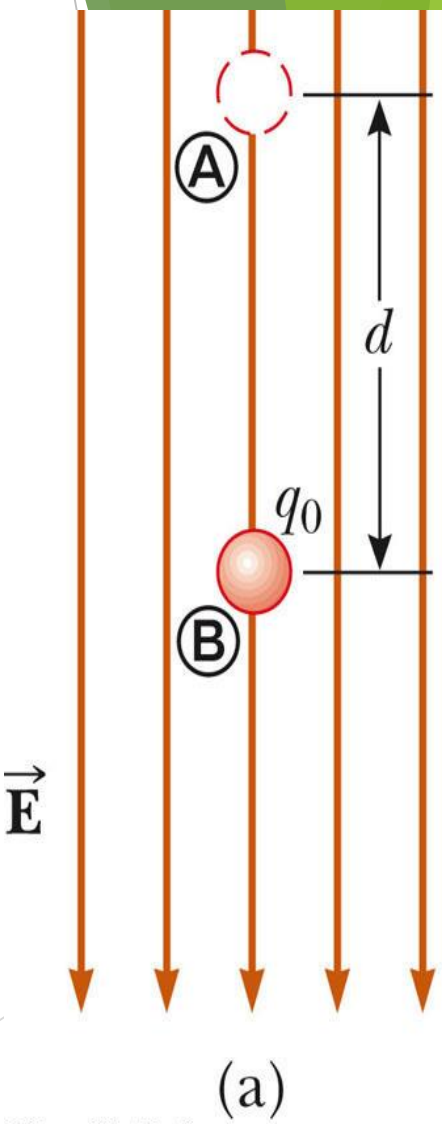
Sec 25.2: Potential Difference in a Uniform Electric Field

- The equations for electric potential can be simplified if the electric field is uniform:

$$V_B - V_A = \Delta V = -Ed$$

- The negative sign indicates that the electric potential at point *B* is lower than at point *A*
  - Electric field lines always point in the direction of decreasing electric potential
- When the electric field is directed downward, point B is at a lower potential than point A
- When a positive test charge moves from A to B, the charge-field system loses potential energy

$$\Delta U = q\,\Delta V = -qEd$$

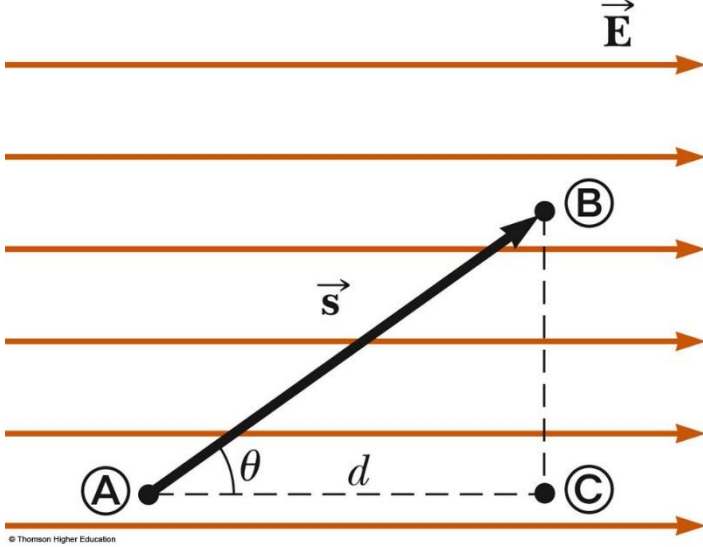


© Thomson Higher Education

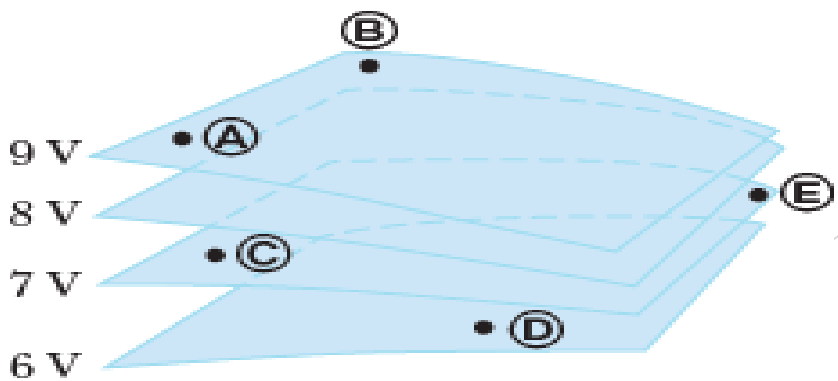
- A system consisting of a positive charge and an electric field **loses** electric potential energy when the charge moves in the direction of the field
  - An electric field does work on a positive charge when the charge moves in the direction of the electric field
- The charged particle gains kinetic energy equal to the potential energy lost by the charge-field system
  - Another example of Conservation of Energy
- If q_o is negative, then ΔU is positive
- A system consisting of a negative charge and an electric field *gains* potential energy when the charge moves in the direction of the field
  - In order for a negative charge to move in the direction of the field, an external agent must do positive work on the charge

Equipotentials

- Point *B* is at a lower potential than point *A*
- Points B and *C* are at the same potential
  - All points in a plane perpendicular to a uniform electric field are at the same electric potential
- The name **equipotential surface** is given to any surface consisting of a continuous distribution of points having the same electric potential

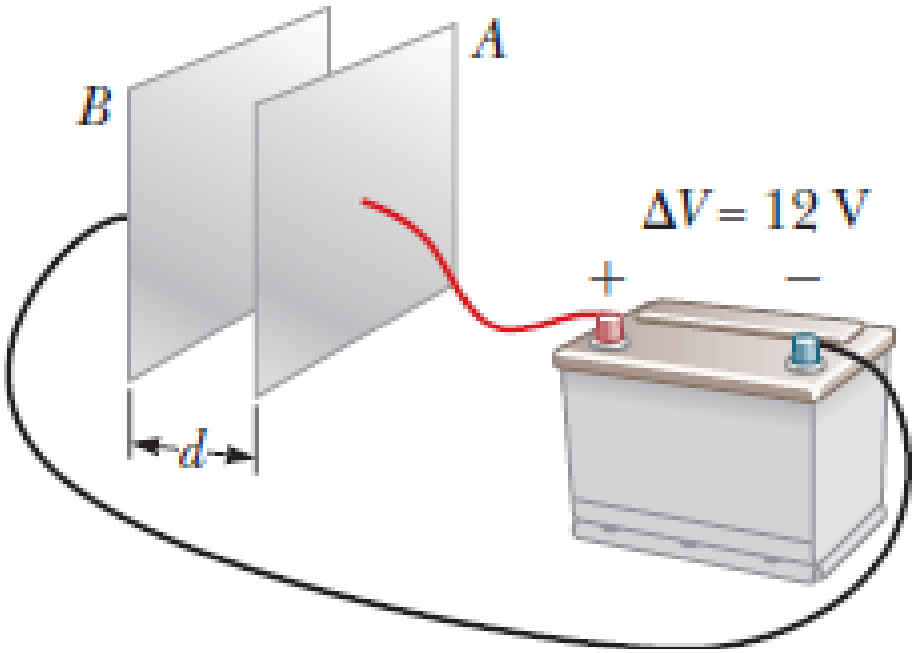


Quick Quiz 25.2 The labeled points in Figure 25.4 are on a series of equipotential surfaces associated with an electric field. Rank (from greatest to least) the work done by the electric field on a positively charged particle that moves from Ⓐ to Ⓑ, from Ⓑ to Ⓒ, from Ⓒ to Ⓓ, and from Ⓓ to Ⓔ.



Example 25.1 The Electric Field Between Two Parallel Plates of Opposite Charge

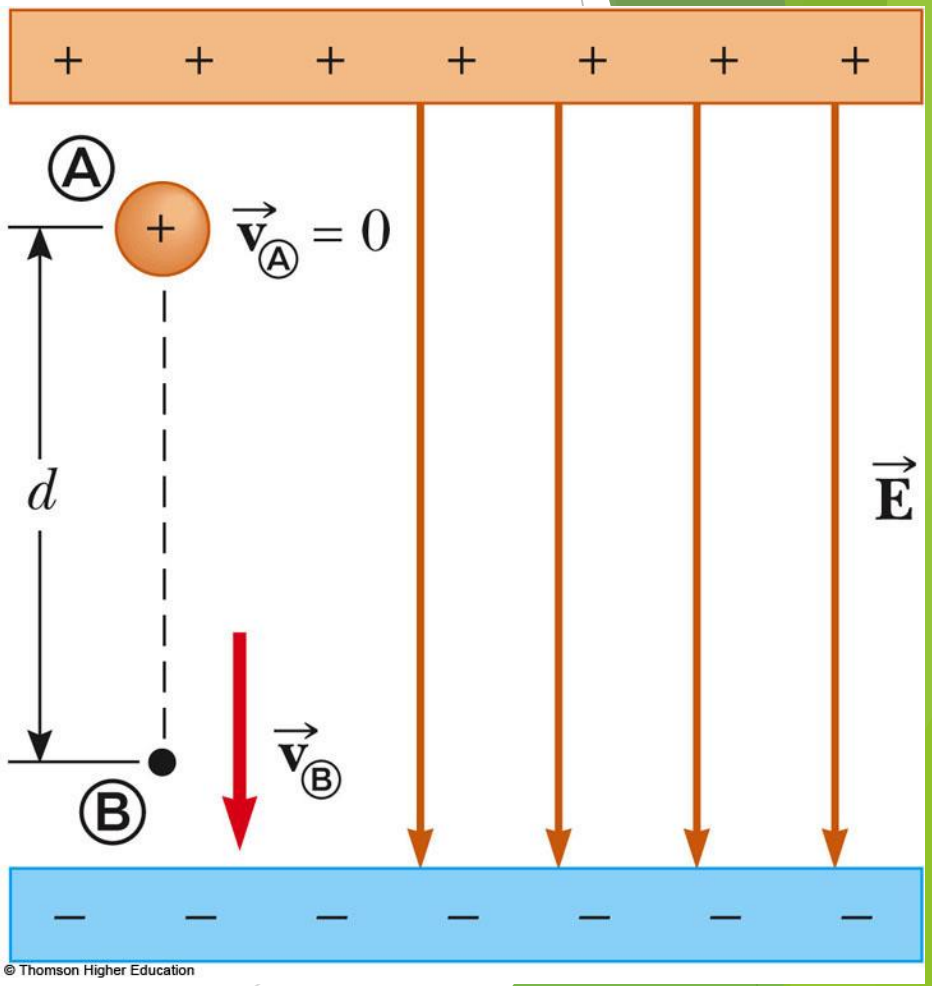
A battery has a specified potential difference ΔV between its terminals and establishes that potential difference between conductors attached to the terminals. A 12-V battery is connected between two parallel plates as shown in Figure 25.5. The separation between the plates is $d = 0.30$ cm, and we assume the electric field between the plates to be uniform. (This assumption is reasonable if the plate separation is small relative to the plate dimensions and we do not consider locations near the plate edges.) Find the magnitude of the electric field between the plates.



$$E = \frac{|V_B - V_A|}{d} = \frac{12\ \text{V}}{0.30 \times 10^{-2}\ \text{m}} = 4.0 \times 10^3\ \text{V/m}$$

Charged Particle in a Uniform Field, Example

- A positive charge is released from rest and moves in the direction of the electric field
- The change in potential is negative
- The change in potential energy is negative
- The force and acceleration are in the direction of the field
- Conservation of Energy can be used to find its speed

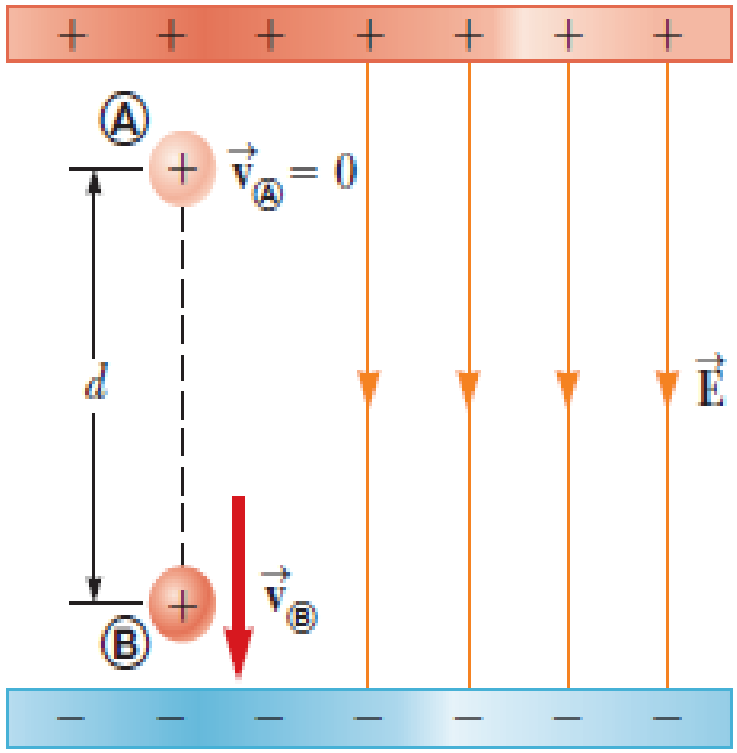


Example 25.2 Motion of a Proton in a Uniform Electric Field

AM

A proton is released from rest at point Ⓐ in a uniform electric field that has a magnitude of 8.0×10^4 V/m (Fig. 25.6). The proton undergoes a displacement of magnitude $d = 0.50$ m to point Ⓑ in the direction of $\vec{\mathbf{E}}$. Find the speed of the proton after completing the displacement.

$$\Delta K + \Delta U = 0$$



$$\left(\tfrac{1}{2}mv^2 - 0\right) + e\,\Delta V = 0$$

$$v = \sqrt{\frac{-2e\,\Delta V}{m}} = \sqrt{\frac{-2e(-Ed)}{m}} = \sqrt{\frac{2eEd}{m}}$$

$$v = \sqrt{\frac{2(1.6 \times 10^{-19}\ \text{C})(8.0 \times 10^4\ \text{V})(0.50\ \text{m})}{1.67 \times 10^{-27}\ \text{kg}}}$$

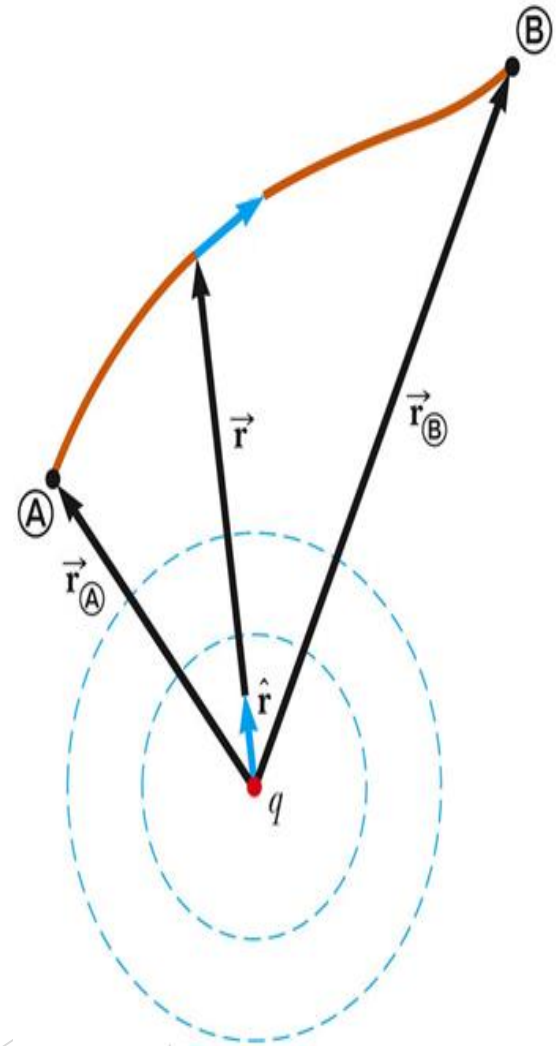
$$= \boxed{2.8 \times 10^6\ \text{m/s}}$$

Sec 25.3: Electric Potential and Potential Energy Due to Point Charges

- A positive point charge produces a field directed radially outward
- The potential difference between points *A* and *B* will be

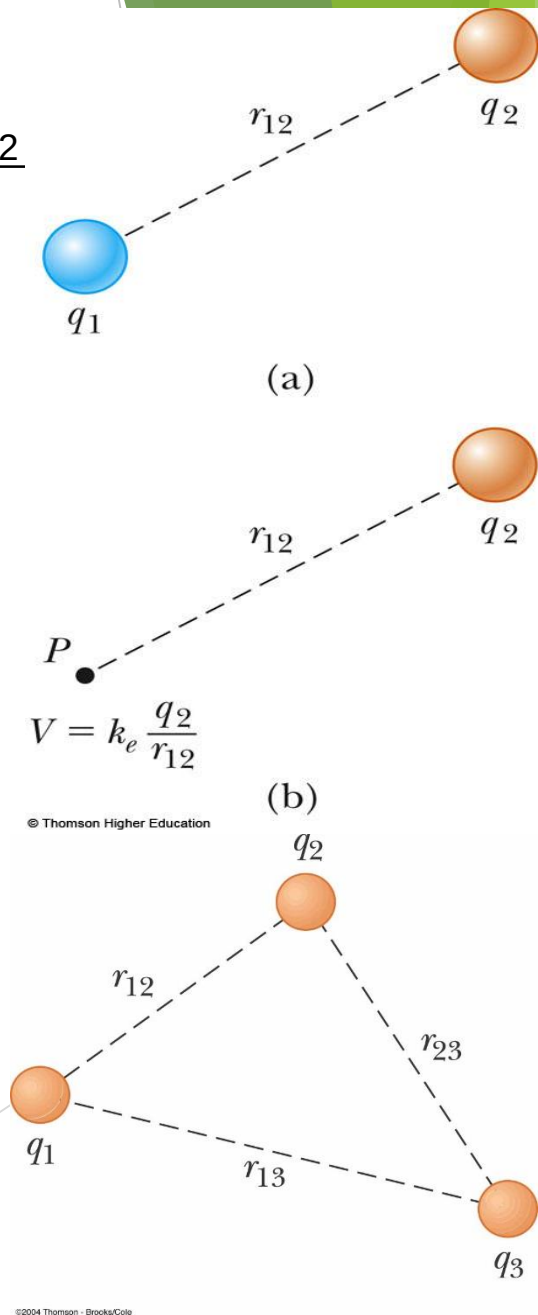
$$V_B - V_A = k_e q\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

- The electric potential is independent of the path between points A and B
- It is customary to choose a reference potential of V = 0 at rA = ∞
- Then the potential at some point r is $V = k_e \frac{q}{r}$
- The electric potential due to several point charges is the sum of the potentials due to each individual charge
  - The sum is the algebraic sum $V = k_e \sum_i \frac{q_i}{r_i}$
    - V = 0 at r = ∞



Potential Energy of Multiple Charges

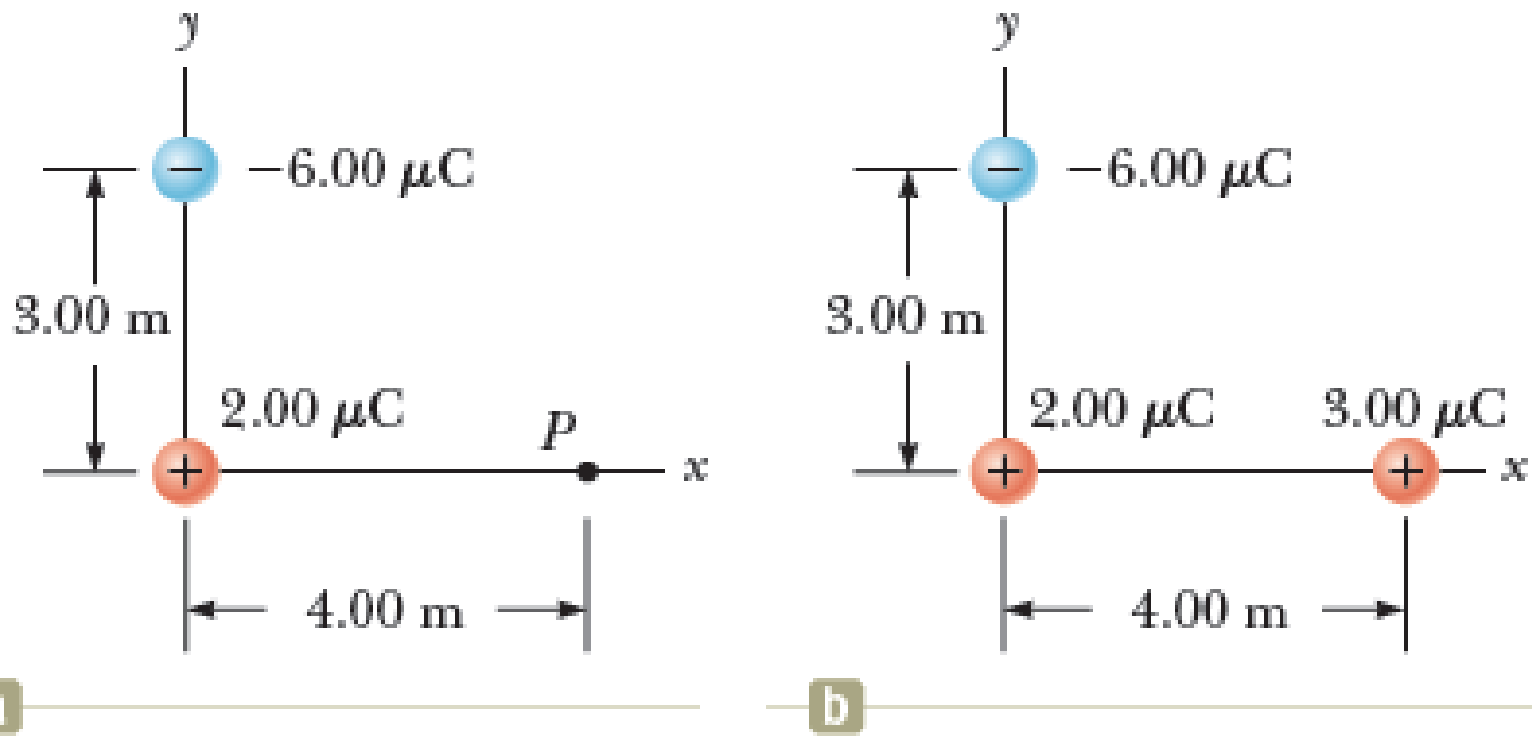
- Consider two charged particles
- The potential energy of the system is $U = k_e \dfrac{q_1 q_2}{r_{12}}$
- If the two charges are the same sign, *U* is positive and work must be done to bring the charges together
- If the two charges have opposite signs, *U* is negative and work is done to keep the charges apart
- If there are more than two charges, then find U for each pair of charges and add them
- For three charges: $U = k_e \left(\dfrac{q_1 q_2}{r_{12}} + \dfrac{q_1 q_3}{r_{13}} + \dfrac{q_2 q_3}{r_{23}} \right)$
  - The result is independent of the order of the charges



Example 25.3 The Electric Potential Due to Two Point Charges

As shown in Figure 25.10a, a charge $q_1 = 2.00\ \mu C$ is located at the origin and a charge $q_2 = -6.00\ \mu C$ is located at (0, 3.00) m.

(A) Find the total electric potential due to these charges at the point P, whose coordinates are (4.00, 0) m.



$$V_P = k_e\left(\frac{q_1}{r_1} + \frac{q_2}{r_2}\right)$$

$$V_P = (8.988 \times 10^9\ \mathrm{N \cdot m^2/C^2})\left(\frac{2.00 \times 10^{-6}\ \mathrm{C}}{4.00\ \mathrm{m}} + \frac{-6.00 \times 10^{-6}\ \mathrm{C}}{5.00\ \mathrm{m}}\right)$$

$$= \boxed{-6.29 \times 10^3\ \mathrm{V}}$$

(B) Find the change in potential energy of the system of two charges plus a third charge $q_3 = 3.00\ \mu\mathrm{C}$ as the latter charge moves from infinity to point P (Fig. 25.10b).

$$U_f = q_3 V_P$$

$$\Delta U = U_f - U_i = q_3 V_P - 0 = (3.00 \times 10^{-6}\ \mathrm{C})(-6.29 \times 10^3\ \mathrm{V})$$

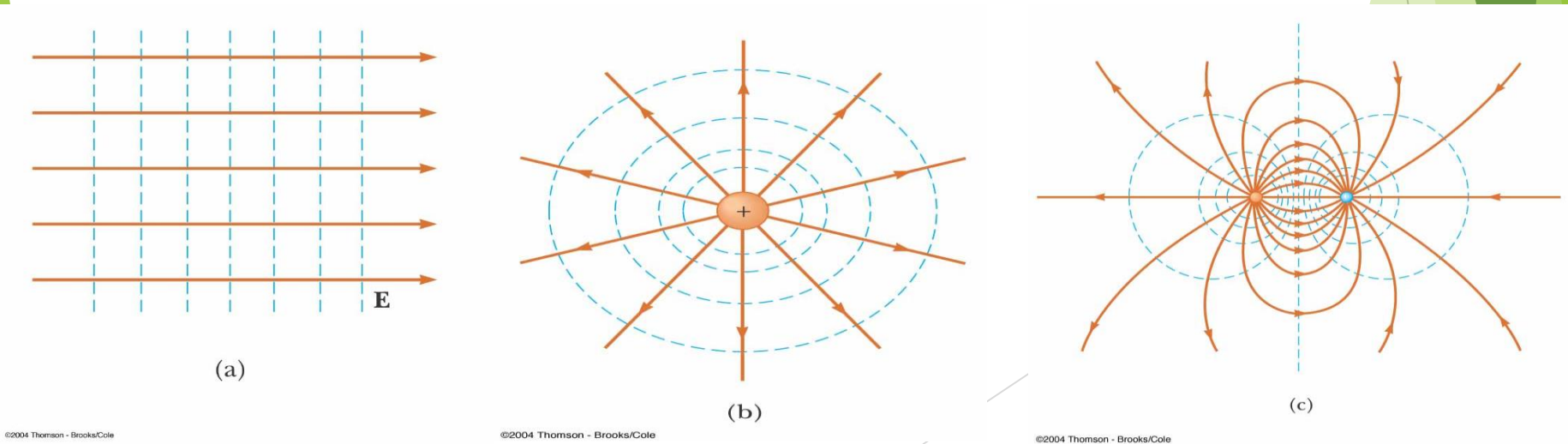
$$= \boxed{-1.89 \times 10^{-2}\ \mathrm{J}}$$

Sec 25.4: Obtaining the Value of the Electric Field from the Electric Potential

- Assume, to start, that the field has only an *x* component $E_x = -\frac{dV}{dx}$
- In general, the electric potential is a function of all three dimensions, given V (x, y, z) you can find Ex, Ey and Ez as partial derivatives

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$

- Equipotential surfaces must always be perpendicular to the electric field lines passing through them
- The equipotential lines are the dashed blue lines, The electric field lines are the brown lines
- The equipotential lines are everywhere perpendicular to the field lines

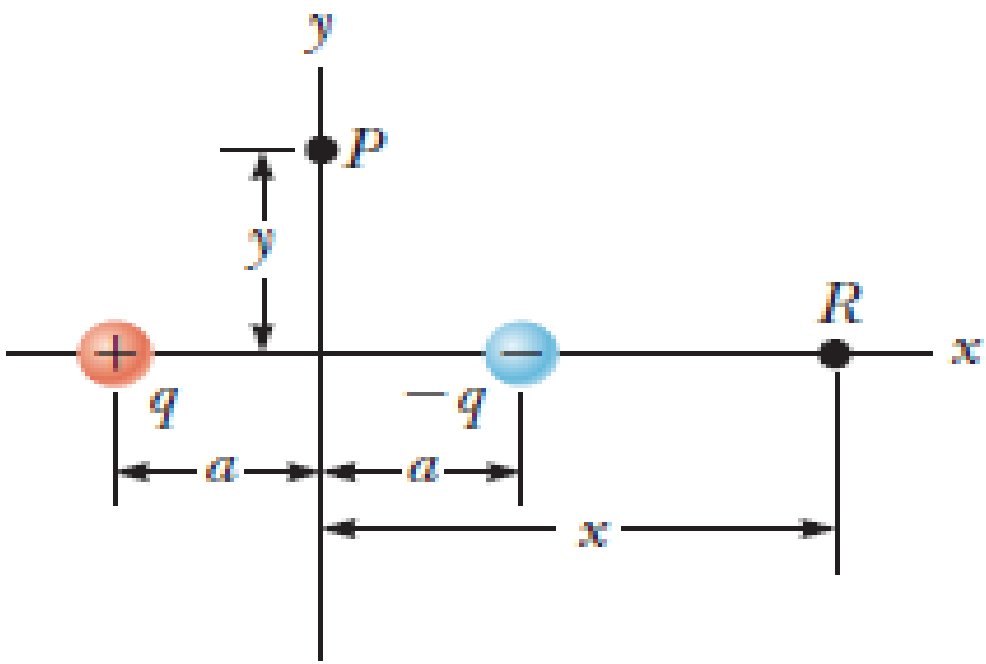


©2004 Thomson - Brooks/Cole
©2004 Thomson - Brooks/Cole
©2004 Thomson - Brooks/Cole

Example 25.4 The Electric Potential Due to a Dipole

An electric dipole consists of two charges of equal magnitude and opposite sign separated by a distance $2a$ as shown in Figure 25.13. The dipole is along the x axis and is centered at the origin.

(A) Calculate the electric potential at point P on the y axis.



$$V_P = k_e \sum_i \frac{q_i}{r_i} = k_e\left(\frac{q}{\sqrt{a^2 + y^2}} + \frac{-q}{\sqrt{a^2 + y^2}}\right) = \boxed{0}$$

(B) Calculate the electric potential at point R on the positive x axis.

$$V_R = k_e \sum_i \frac{q_i}{r_i} = k_e \left(\frac{-q}{x-a} + \frac{q}{x+a} \right) = -\frac{2k_e qa}{x^2 - a^2}$$

(C) Calculate V and E_x at a point on the x axis far from the dipole.

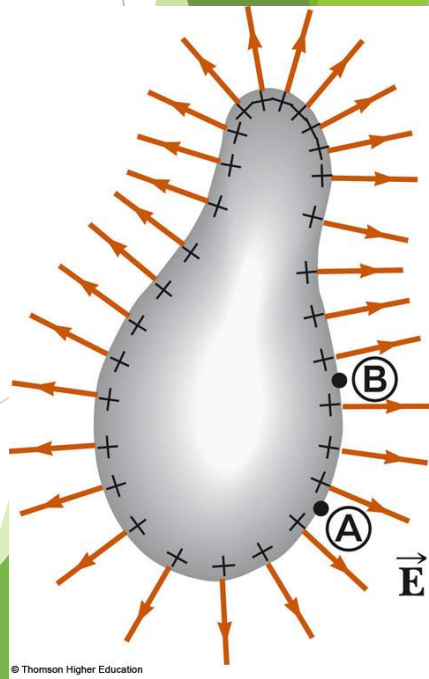
$$V_R = \lim_{x>>a} \left(-\frac{2k_e qa}{x^2 - a^2} \right) \approx -\frac{2k_e qa}{x^2} \quad (x >> a)$$

$$E_x = -\frac{dV}{dx} = -\frac{d}{dx}\left(-\frac{2k_e qa}{x^2} \right)$$

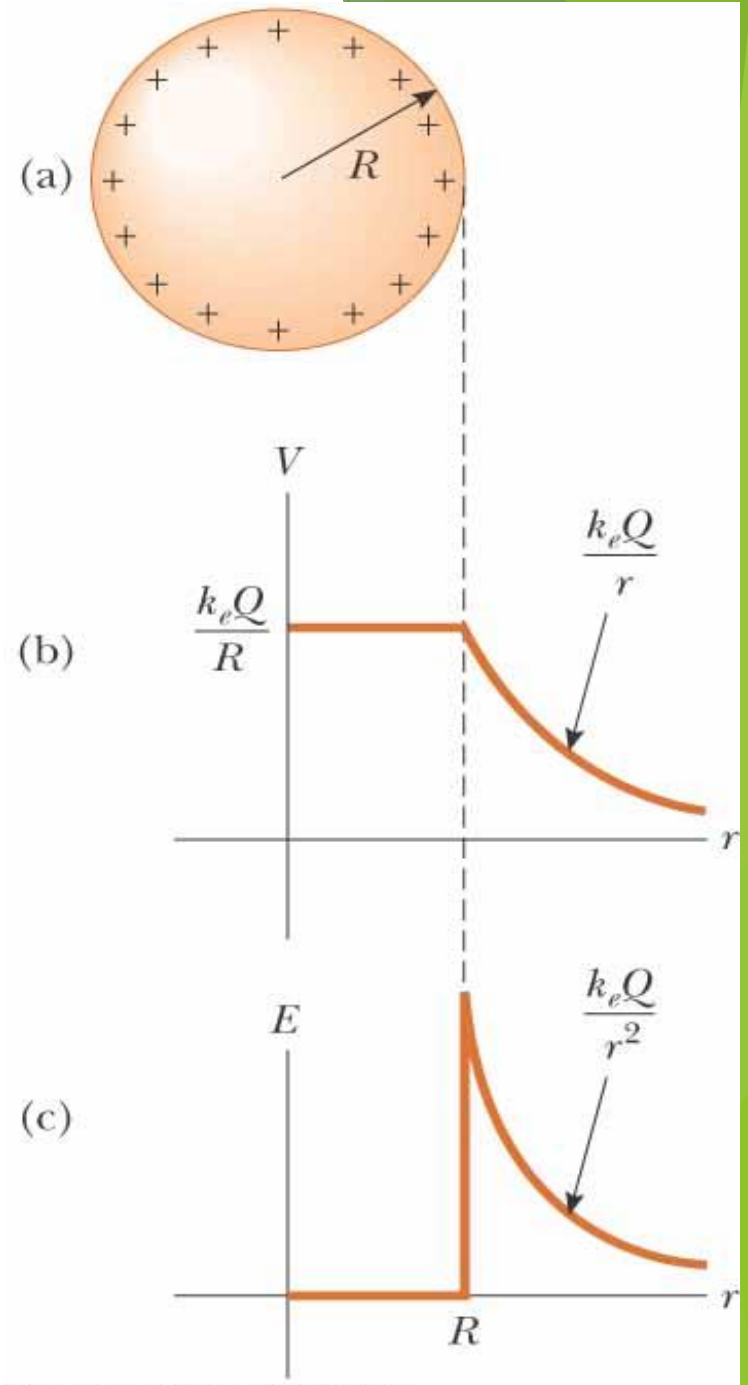
$$= 2k_e qa \frac{d}{dx}\left(\frac{1}{x^2} \right) = -\frac{4k_e qa}{x^3} \quad (x >> a)$$

Sec 25.6: Electric Potential Due to a Charged Conductor

- potential difference between *A* and *B* is zero
- *V* is constant everywhere on the surface of a charged conductor in equilibrium
  - $\Delta V = 0$ between any two points on the surface
- The surface of any charged conductor in electrostatic equilibrium is an equipotential surface
- Because the electric field is zero inside the conductor, we conclude that the electric potential is constant everywhere inside the conductor and equal to the value at the surface



E Compared to *V*



©2004 Thomson - Brooks/Cole

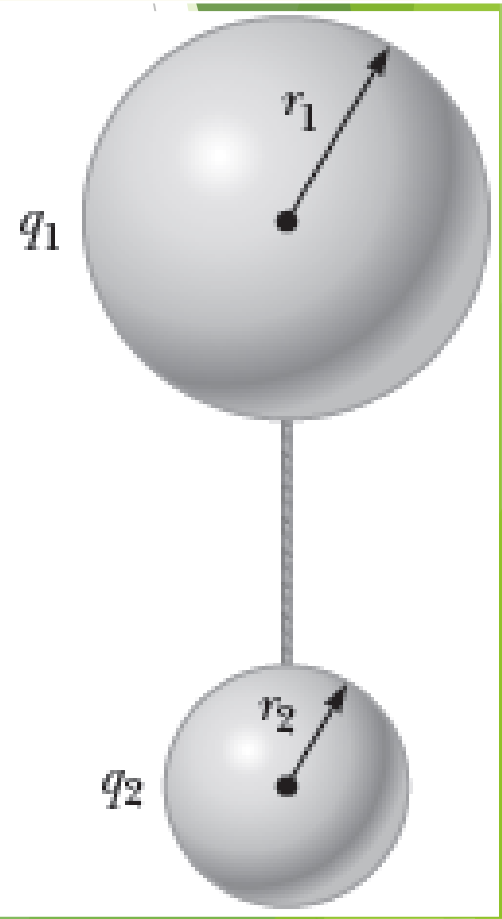
Example 25.8 Two Connected Charged Spheres

Two spherical conductors of radii r_1 and r_2 are separated by a distance much greater than the radius of either sphere. The spheres are connected by a conducting wire as shown in Figure 25.19. The charges on the spheres in equilibrium are q_1 and q_2, respectively, and they are uniformly charged. Find the ratio of the magnitudes of the electric fields at the surfaces of the spheres.

$$V = k_e \frac{q_1}{r_1} = k_e \frac{q_2}{r_2}$$

$$\frac{q_1}{q_2} = \frac{r_1}{r_2}$$

$$E_1 = k_e \frac{q_1}{r_1^2} \quad \text{and} \quad E_2 = k_e \frac{q_2}{r_2^2}$$

$$\frac{E_1}{E_2} = \frac{q_1}{q_2} \frac{r_2^2}{r_1^2}$$

$$\frac{E_1}{E_2} = \frac{r_1}{r_2} \frac{r_2^2}{r_1^2} = \boxed{\frac{r_2}{r_1}}$$